Assignment 1: Group 4

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TASK 1

1. Total Rows: 54,960,961

2. Unique IDs: 1,430,445

3. Time Range: 2020-11-03 03:31:30+00:00 (November 3rd, 2020 @ 3:30UTC) to 2023-01-23 12:29:56+00:00 (January 23rd, 2023 @ 12:29UTC)

TASK 2

1. Total duplicates found: 3,112,100

2. Proportion of duplicates: 5.66%

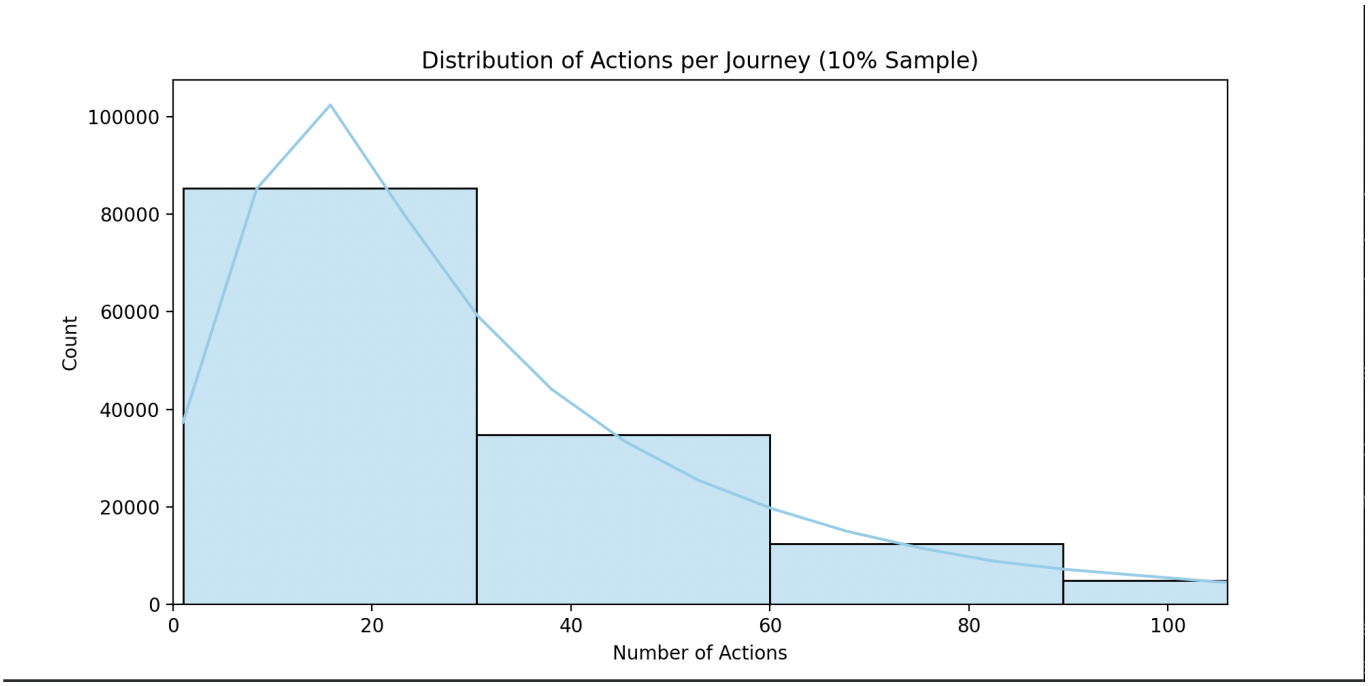
3. Rows remaining after cleaning: 51,848,861

TASK 3

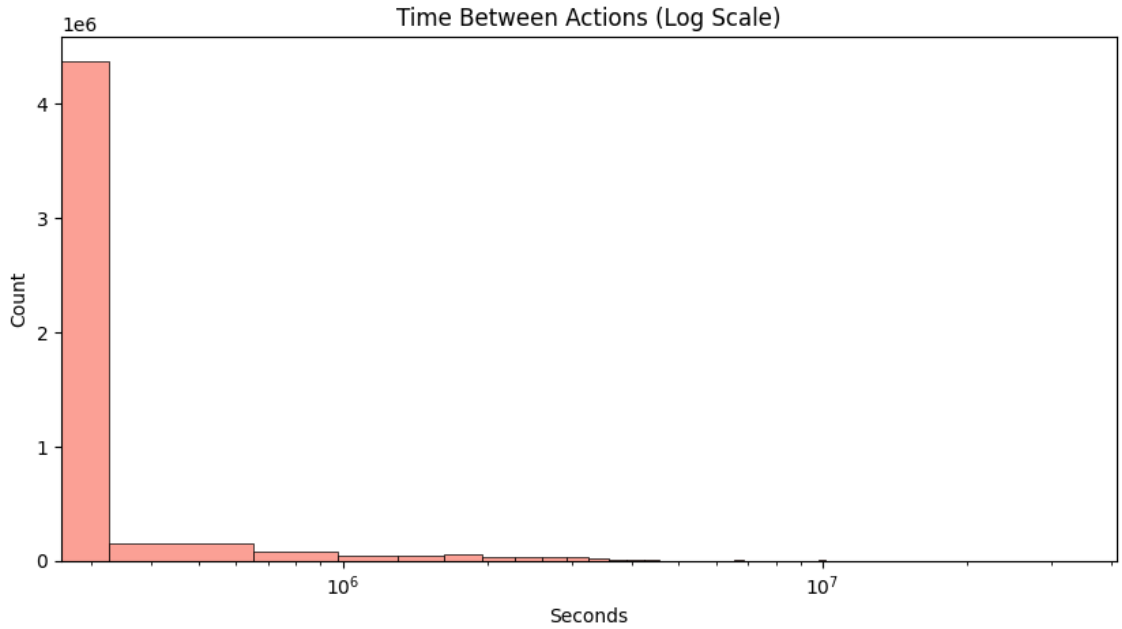
I sampled 10% of the data on seed(42), to get 143,000 users, then proceeded with summary stats I used median over mean in the case of large outliers (bots) which would skew results

1. Typical Journey Characteristics

Median Actions per Journey: 24.0 actions



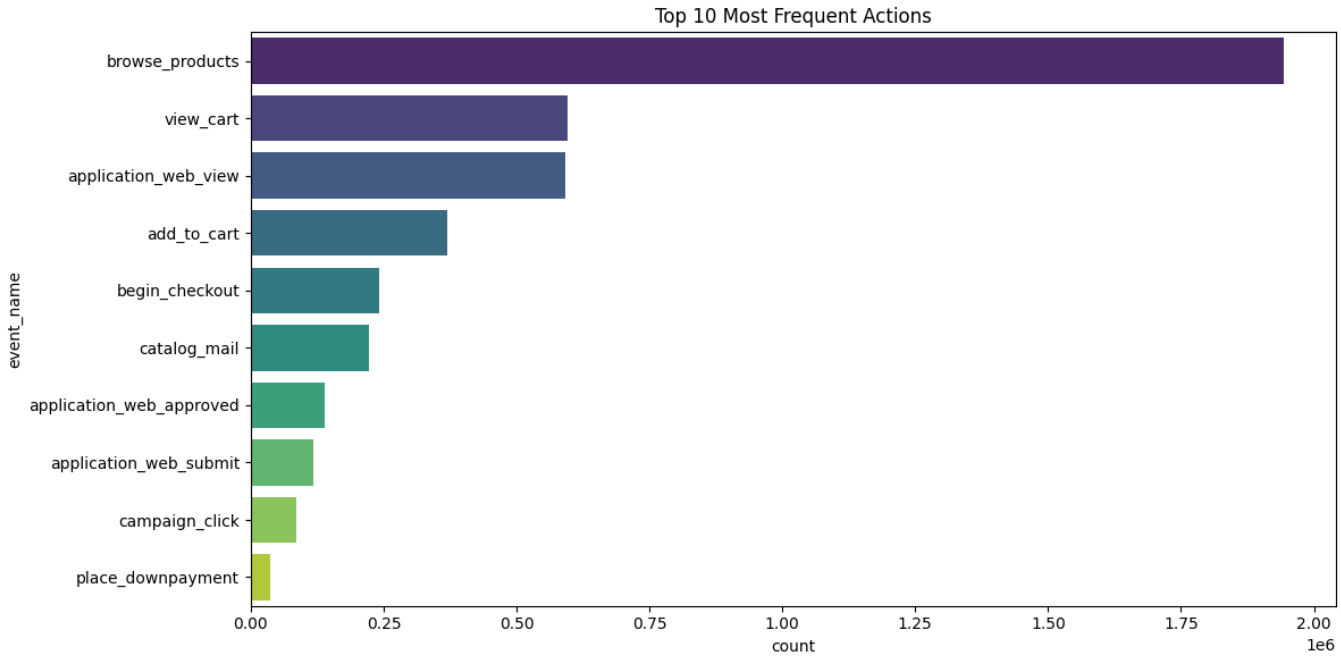
Median Journey Duration: 140,758.61 minutes (~97.75 days) **Median between Actions:** 88 seconds (Median time between actions)



Insight: While the total "life" of a user ID in the system spans roughly 3 months, the 88-second median between actions suggests that actual engagement occurs in smaller sessions.

2. Common User Actions

Event Name	Total Occurrences	Stage
Browse Products	1,944,325	First Purchase
View Cart	595,620	First Purchase
Application Web View	590,662	Apply for Credit



3. Success Case Analysis

A "Success" is defined as any journey containing at least one **order_shipped** event (Event ID: 28). This segment represents **19.53%** (279,363 users) of the total population.

The following table highlights the difference in behavior between users who convert and those who do not.

Metric	Non-Successful Users (80.47%)	Successful Users (19.53%)
Median Actions	21.0	**41.0**
Average Actions	31.0	**54.0**
Median Duration	159.0 Days	**18.0 Days**
Average Duration	154.0 Days	**48.0 Days**

Key findings

1. **Intensity of Engagement:** Successful users are nearly twice as active (41 vs 21 median actions) as those who do not purchase.
2. **The "Conversion Window":** Successful users convert relatively quickly (Median: **18 days**), whereas non-successful users tend to linger in the "browsing" state for over **154 dayss**
3. **Progression:** The high frequency of "Application Web View" in the apply for credit stage suggests that navigating the credit application is a standard prerequisite for the successful "Order Shipped" outcome.